FEUP

FACULTY OF ENGINEERING - UNIVERSITY OF PORTO

Systems Engineering

System Concept

SWAID - Plate Resonance (PR)

PLATE RESONANCE
TEAM

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1 Introduction

1.1 Document Scope

This document presents Deliverable #5 for the Systems Engineering course and outlines the System Concept for the Plate Resonance (PR) subsystem, developed by Group 13 as part of the Standing Waves Acoustic Interactive Demonstrator (SWAID) project. The scope of this deliverable is strictly aligned with defining the foundational structure and operational logic of the PR subsystem.

1.2 Acronyms and Terms

SWAID Standing Wave Acoustic Interactive Demonstrator.

PR Plate Resonance subsystem of SWAID.

COTS Commercial Off-The-Shelf

HI Human Interface

1.3 Document Revision

Table 1: Revision history

Version	Date	Author(s)	Description
1.0	2026-02-27	Lucas Aparício	Template creation
2.0	2026-03-27	João Pinto	First Version
3.0	2026-03-27	Hugo Gomes / Alexandre Gorra	Final Version

2 System Overview

2.1 Overview of the Challenge

The SWAID project aims to visually demonstrate acoustic standing waves through user interaction. Within this ecosystem, the primary challenge of the PR subsystem is to accurately generate and synchronize multiple audio-frequency signals that will be fed into the external amplification system, alongside controlling a lighting system synchronized with the acoustic signals.

The system must dynamically adjust frequencies, amplitudes, and phases in near real-time based on modulation commands received from the Human Interface (HI) subsystem. The core engineering challenge lies in ensuring high signal fidelity, minimal latency, and perfect phase synchronization between multiple output channels to guarantee stable geometric patterns on the plate. To bypass the complexity and development time of custom FPGA hardware design, this subsystem innovates by utilizing a Commercial Off-The-Shelf (COTS) multichannel audio interface, shifting the synthesis challenge from hardware to software while maintaining strict acoustic determinism.

2.2 System Breakdown Structure

The System Breakdown Structure below organizes the PR subsystem's scope into three primary functional pillars. It strictly encompasses the pipeline from receiving HI commands to generating the analog audio signals and controlling the lighting effects, explicitly excluding external amplification and physical actuation.

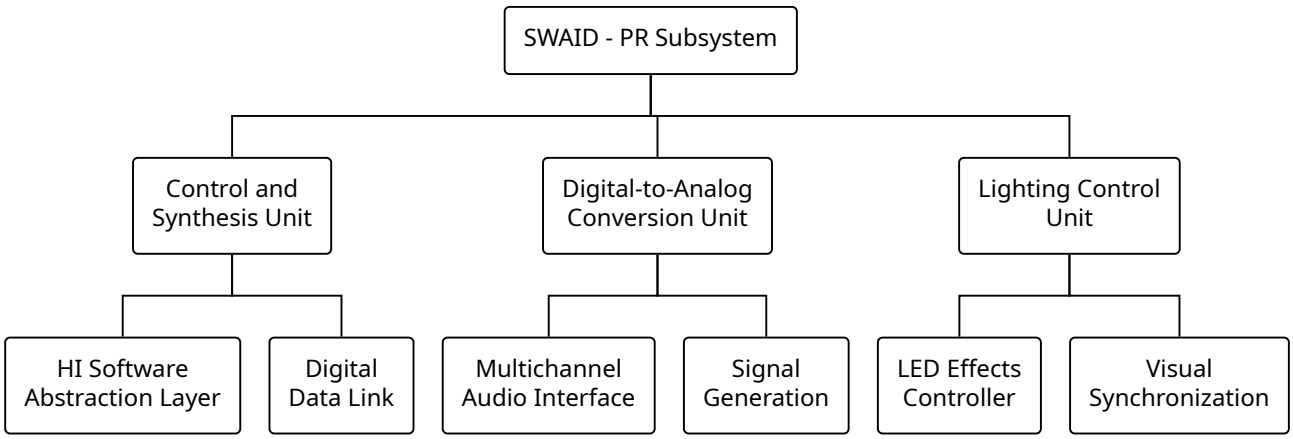

Figure 1: System Breakdown Structure (SBS) - SWAID Project

2.3 Subsystem Component Descriptions

To clarify the specific responsibilities within the PR subsystem, the functional blocks identified in the System Breakdown Structure are detailed below:

1. Control and Synthesis Unit

- **HI Software Abstraction Layer**: Represents the software library that permits the Human Interface team to send high-level digital commands (specifying target Chladni symbols,

LED illumination effects, and volume controls). It translates these requests through a lookup table into the low-level parameters required by the downstream units to produce their respective physical effects.

- **Digital Data Link**: Represents the communication protocols (e.g., USB, Serial) responsible for the robust transfer of information between the main computational unit (PC) and the external hardware peripherals.

2. Digital-to-Analog Conversion Unit

- **Signal Generation Software**: The software and drivers that handle the actual signal synthesis and routing, ensuring that the translated acoustic parameters are correctly formatted for the hardware.
- **Multichannel Audio Interface**: The physical sound card that receives the digital audio streams and converts them into precise, multi-channel analog signals ready for external amplification.

3. Lighting Control Unit

- **LED Effects Controller**: The dedicated hardware (microcontroller) responsible for driving the physical LED strips, executing the specific illumination IDs requested by the abstraction layer.
- **Visual Synchronization**: The internal logic that ensures the lighting transitions, cross-fades, and rhythmic pulses occur in perfect phase alignment with the acoustic pattern changes dictated by the Transition Time parameter.

3 Annex

3.1 Engineering Requirements

Table 2: SWAID Plate Resonance (PR) Requirements Specification

ID	Type	Priority	Description
Plate Resonance (PR) Subsystem			
PR-01	Functional	High	The PR subsystem shall generate independent, synchronized actuation signals, specifying frequency, phase, and amplitude, for the acoustic transducers.
PR-02	Functional	High	The PR subsystem shall maintain an internal mapping library to directly translate high-level target pattern commands into precise transducer actuation parameters.
PR-03	Functional	Medium	The PR subsystem shall output a state-change signal to the lighting controller whenever the active resonance mode changes.
PR-04	Performance	High	The PR subsystem processing latency, measured from the receipt of an HI command to the output of transducer actuation signals, shall not exceed 20 milliseconds.
PR-05	Performance	Medium	The PR subsystem shall ensure strict temporal synchronization between acoustic actuation signals and lighting control commands, with jitter not exceeding 10 milliseconds.
PR-06	Quality Attribute	Medium	The PR signal generation architecture shall be abstracted (alphabet-agnostic) to support dynamic remapping and expansion of the resonance library without modifying core generation logic.
PR-07	Functional	High	The PR subsystem firmware shall enforce digital limits on maximum signal amplitude and amplifiers gain to prevent thermal overloading and damage to the CP transducers.
PR-08	Design Constraint	Medium	The PR subsystem processing hardware shall conform to the dimensional constraints of the electronics compartment designated by the CP subsystem.
PR-09	Internal Interface	High	The PR subsystem shall expose a software API to accept standard high-level command strings (Chladni Symbols, Illumination effects, Symbol Transition Timing) from the HI subsystem.
PR-10	Internal Interface	High	The PR subsystem shall transmit analog actuation signals to the CP transducer drivers via a standard stereo audio jack connection.
PR-11	Performance	Low	The PR subsystem shall initialize, load the resonance dictionary into memory, and achieve a ready state within 10 seconds of receiving power.

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Table 2 – Continued from previous page

ID	Type	Priority	Description
PR-12	Functional	High	The PR subsystem resonance mode library shall contain a minimum of 12 mathematically distinct and repeatable Chladni patterns to map to a full musical octave.

3.2 Market Survey

Table 3: Market Survey

System Type	Signal Gen.	Channels	Latency	Complexity	Technical Note
Traditional Demonstrator	Analog Gen.	1	N/A	Low	Limited to single-frequency study; no multi-point control.
Audio Interface	**Sound Card**	**6–8**	**Low***	**Low**	**Selected: 5.1/7.1 is sufficient for 4–6 transducers via ALSA.**
FPGA Generator	DDS Logic	8+	Ultra-Low	High	Over-engineered for plate size; high development time/cost.
Microcontroller	PWM / DAC	1–2	Moderate	Mod/High	Difficult to synchronize phase,amplitude and frequency across multiple channels.

* *Low latency achieved through direct ALSA driver optimization.*

3.3 Functional Architecture

The functional architecture of the Plate Resonance (PR) subsystem is centered around the real-time translation of high-level commands from the Human Interface (HI) subsystem into synchronized analog signals for both acoustic excitation and lighting control. Importantly, the PR subsystem's scope ends at the generation of these analog signals—amplification and acoustic actuation are handled by downstream hardware outside the subsystem boundary.

The architecture is structured around three core input parameters received from the HI team:

- **Symbol ID**: Refers to a lookup table containing the exact parameters (amplitude, phase, and frequency) required for each independent transducer to generate the target pattern.
- **LED ID**: Triggers a pre-programmed lighting effect that is perfectly synchronized with the selected acoustic pattern.
- **Transition Time**: A parameter that controls the speed of the crossfade or switch between the current symbol and the next, enabling dynamic visual flow and the representation of rhythm.
- **Volume**: A parameter that controls the gain applied to the transducer amplifiers, allowing dynamic adjustment of the overall acoustic intensity.

These inputs are processed by the PR subsystem's software layer, which interfaces with a Commercial Off-The-Shelf (COTS) multichannel audio interface. The audio interface generates the necessary low-level analog signals that are then sent to the external amplification and transducer units. Lighting control signals are generated in parallel, ensuring tight synchronization with the acoustic output.

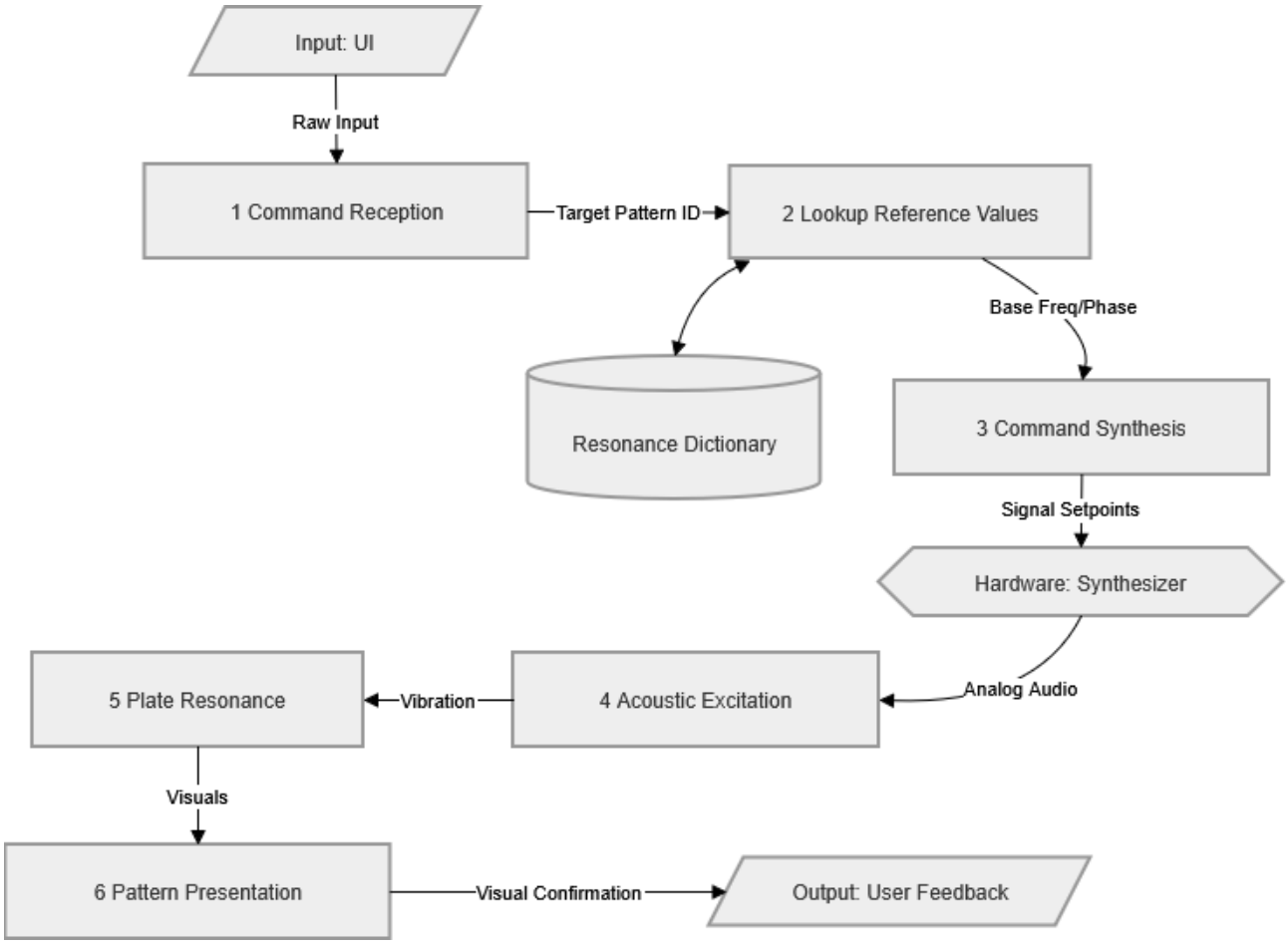

Figure 2: Functional architecture diagram.

4 Conclusion

In conclusion, we were able to establish the foundational analysis and functional architecture required to develop the SWAID Plate Resonance subsystem. By defining a set of objectives and technical requirements, clear boundaries have been set for system performance,like the latency limits and critical hardware safety constraints.

With the theoretical framework, interface architectures, and hardware strategies now solidified, the team is fully prepared to advance into the next phases.